

Partially inelastic surface — Solution

X22 (2022) — English translation

A1

0.10 pts

Express $\dot{v}_x$ in terms of F_x and m .

From the theorem on the motion of the center of mass

$$F_x = m\dot{v}_x$$

from which

$$\dot{v}_x = \frac{F_x}{m}$$

Answer: Answer:

$$\dot{v}_x = \frac{F_x}{m}$$

A2

0.40 pts

Express $\dot{\omega}$ in terms of F_x , N , L , φ and m .

From the equation of dynamics of rotational motion relative to the center of mass

$$I_O \dot{\omega} = \frac{mL^2 \dot{\omega}}{3} = NL \sin \varphi - F_x L \cos \varphi$$

from which

$$\dot{\omega} = \frac{3}{mL} (N \sin \varphi - F_x \cos \varphi)$$

Answer: Answer:

$$\dot{\omega} = \frac{3}{mL} (N \sin \varphi - F_x \cos \varphi)$$

A3

0.10 pts

Express v_{Ax} in terms of ω , v_x , φ and L .

From the theorem on the addition of velocities

$$\vec{v}_A = \vec{v}_O + \vec{v}_{\text{rel}}$$

from which

$$v_{Ax} = v_x - \omega L \cos \varphi$$

Answer: Answer:

$$v_{Ax} = v_x - \omega L \cos \varphi$$

A4

0.40 pts

Prove that $v_{Ax} \leq 0$ for any values of μ and φ .

If $v_{Ax} > 0$, then $F_x = -\mu N$ Then

$$\omega L = -\frac{3v_x}{\mu} (\sin \varphi + \mu \cos \varphi)$$

or same

$$v_{A_x} = v_x \left(1 + \frac{3 \cos \varphi}{\mu} (\sin \varphi + \mu \cos \varphi) \right)$$

But at the same time $v_x < 0$, therefore $v_{A_x} < 0$, which leads to a contradiction.

A5

0.10 pts

Express v_{A_x} in terms of v_x , φ and μ .

In this case $F_x = \mu N$. Then

$$\omega L = \frac{3v_x}{\mu} (\sin \varphi - \mu \cos \varphi)$$

or same

$$v_{A_x} = v_x \left(1 - \frac{3 \cos \varphi}{\mu} (\sin \varphi - \mu \cos \varphi) \right)$$

Answer: Answer:

$$v_{A_x} = v_x \left(1 - \frac{3 \cos \varphi}{\mu} (\sin \varphi - \mu \cos \varphi) \right)$$

A6

0.60 pts

At what minimum possible value μ_0 is slippage impossible for any values of angle φ ?

Considering that

$$\begin{aligned} \sin 2\varphi &= 2 \sin \varphi \cos \varphi \\ \cos^2 \varphi &= \frac{1 + \cos 2\varphi}{2} \end{aligned}$$

after несложных transformation

$$v_{A_x} = -\frac{v_x}{2\mu} (3 \sin 2\varphi - 3\mu \cos 2\varphi - 5\mu)$$

Introducing variable $\varphi_0 = \arctan \mu$

$$v_{A_x} = -\frac{v_x}{2\mu} \left(3\sqrt{1 + \mu^2} \sin(2\varphi - \varphi_0) - 5\mu \right)$$

Condition $v_{A_x} < 0$ реализуется then when expression in bracket положительно. Since $\sin(2\varphi - \varphi_0) \leq 1$

$$\frac{5\mu}{3\sqrt{1 + \mu^2}} \leq 1$$

or same

$$\mu_0 = \frac{3}{4}$$

second method

For наличии проскальзывания $F_x = \mu N$. Then can express μ through φ as

$$\mu = \frac{3 \cos \varphi \sin \varphi}{1 + 3 \cos^2 \varphi}$$

Find derivative

$$\frac{d\mu}{d\varphi} = \frac{3}{1 + 3 \cos^2 \varphi} \left((\cos^2 \varphi - \sin^2 \varphi)(1 + 3 \cos^2 \varphi) + 6 \sin^2 \varphi \cos^2 \varphi \right)$$

from which find point экстремума

$$\varphi = \arctan(2)$$

and correspond maximum value

$$\mu_0 = \frac{3}{4}$$

Answer: Answer:

$$\mu_0 = \frac{3}{4}$$

A7

0.80 pts

For $\mu < \mu_0$, find all values of the angle φ at which slippage is possible. Express the boundaries of the range of angles using μ .

The condition for the positive value in parentheses is

$$\arcsin \frac{5\mu}{3\sqrt{1+\mu^2}} \leq 2\varphi - \varphi_0 \leq \pi - \arcsin \frac{5\mu}{3\sqrt{1+\mu^2}}$$

from which

$$\frac{\arcsin \frac{5\mu}{3\sqrt{1+\mu^2}} + \arctan \mu}{2} \leq \varphi \leq \frac{\pi - \arcsin \frac{5\mu}{3\sqrt{1+\mu^2}} + \arctan \mu}{2}$$

second method

Solve equation

$$\mu = \frac{3 \cos \varphi \sin \varphi}{1 + 3 \cos^2 \varphi}$$

as a result transformation which obtain two root

$$\varphi_1 = \arctan \frac{3 - \sqrt{9 - 16\mu^2}}{2\mu}, \quad \varphi_2 = \arctan \frac{3 + \sqrt{9 - 16\mu^2}}{2\mu}$$

from which

$$\arctan \frac{3 - \sqrt{9 - 16\mu^2}}{2\mu} \leq \varphi \leq \arctan \frac{3 + \sqrt{9 - 16\mu^2}}{2\mu}$$

which is identical to the answer obtained by the first method.

Answer: Answer:

$$\arctan \frac{3 - \sqrt{9 - 16\mu^2}}{2\mu} \leq \varphi \leq \arctan \frac{3 + \sqrt{9 - 16\mu^2}}{2\mu}$$

B1

0.20 pts

Express v_x in terms of ω , L and φ .

In this case, the speed of point A is directed vertically, so

$$v_x = \omega L \cos \varphi$$

Answer: Answer:

$$v_x = \omega L \cos \varphi$$

B2

0.60 pts

Express v_y in terms of v_0 , ω , L and φ .

The angular momentum of the rod relative to point A upon impact is conserved. The expression for it is

$$\vec{L}_A = \vec{L}_O + m[\vec{A}\vec{O} \times \vec{v}_O] = \frac{mL^2\vec{\omega}}{3} + m[\vec{A}\vec{O} \times \vec{v}_O]$$

Thus

$$mLv_0 \sin \varphi = mLv_y \sin \varphi + mLv_x \cos \varphi + \frac{mL^2\omega}{3}$$

Account for relation v_x and ω , find in previous part

$$v_y = v_0 - \frac{\omega L(1 + 3 \cos^2 \varphi)}{3 \sin \varphi}$$

Answer: Answer:

$$v_y = v_0 - \frac{\omega L(1 + 3 \cos^2 \varphi)}{3 \sin \varphi}$$

B3

0.80 pts

Find the maximum energy W spent on surface deformation. Express your answer in terms of m , v_0 and φ .

For v_{Ay} we have

$$v_{Ay} = v_y - \omega L \sin \varphi = v_0 - \frac{4\omega L}{3 \sin \varphi}$$

For energy deformation surface E_N have

$$E_N = \int N v_{Ay} dt$$

But from theorem o motion of the center of mass and part B2 have

$$N = -m\dot{v}_y = \frac{m\dot{\omega}L(1 + 3 \cos^2 \varphi)}{3 \sin \varphi}$$

Then

$$E_N(\omega) = \int_0^\omega \frac{m(1 + 3 \cos^2 \varphi)}{3 \sin \varphi} \left(v_0 - \frac{4\omega L}{3 \sin \varphi} \right) L \dot{\omega} dt = \frac{m(1 + 3 \cos^2 \varphi)}{3 \sin \varphi} \left(v_0 \omega L - \frac{2\omega^2 L^2}{3 \sin \varphi} \right)$$

Maximum value W be achieved for $v_{Ay} = 0$, thus.be in this moment □13

Answer: Answer:

$$W = \frac{mv_0^2(1 + 3 \cos^2 \varphi)}{8}$$

B4

1.20 pts

Find ω_{to} after the hit. Express your answer in terms of v_0 , L , φ and k .

At the end of the strike

$$E_N = E_{\text{def}} = W(1 - k) = \frac{mv_0^2(1 + 3 \cos^2 \varphi)(1 - k)}{8}$$

from which

$$\frac{m(1 + 3 \cos^2 \varphi)}{3 \sin \varphi} \left(v_0 \omega L - \frac{2\omega^2 L^2}{3 \sin \varphi} \right) = \frac{mv_0^2(1 + 3 \cos^2 \varphi)(1 - k)}{8}$$

or

$$\omega^2 L^2 - \frac{3v_0 \omega L \sin \varphi}{2} + \frac{9v_0^2 \sin^2 \varphi (1 - k)}{16}$$

Solve square equation

$$\omega = \frac{3v_0 \sin \varphi (1 \pm \sqrt{k})}{4L}$$

Necessarily choose larger from root. Thus

$$\omega_{to} = \frac{3v_0 \sin \varphi (1 + \sqrt{k})}{4L}$$

Answer: Answer:

$$\omega_{to} = \frac{3v_0 \sin \varphi (1 + \sqrt{k})}{4L}$$

B5

0.30 pts

Find the velocity of point A of rod v_{Ato} immediately after the impact. Express your answer using v_0 , φ and k .

$$v_{Ay} = v_0 - \frac{4\omega_{to}L}{3 \sin \varphi} = -v_0 \sqrt{k}$$

or

$$v_{Ato} = v_0 \sqrt{k}$$

Answer: Answer:

$$v_{Ato} = v_0 \sqrt{k}$$

B6

0.40 pts

At what distance L_{AC} from point A is point C of the rod, the speed of which is minimal immediately after the impact? Express your answer using L , φ and k .

Let's find a point that does not have a velocity component, directed perpendicular to the rod, its speed will be minimal

$$L_{AC} = \frac{v_{Ato} \sin \varphi}{\omega_{to}} = L \frac{4\sqrt{k}}{3(1 + \sqrt{k})}$$

Answer: Answer:

$$L_{AC} = L \frac{4\sqrt{k}}{3(1 + \sqrt{k})}$$

C1

0.40 pts

Express ω and v_y in terms of v_x , v_0 , L , μ and φ .

ω was found earlier

$$\omega = \frac{3v_x}{\mu L} (\sin \varphi - \mu \cos \varphi)$$

From theorem o motion of the center of mass

$$m\dot{v}_y = -N$$

from which

$$v_y = v_0 - \frac{v_x}{\mu}$$

Answer: Answer:

$$\omega = \frac{3v_x}{\mu L} (\sin \varphi - \mu \cos \varphi), v_y = v_0 - \frac{v_x}{\mu}$$

C2

1.20 pts

Find the maximum energy W spent on surface deformation. Express your answer in terms of m , v_0 , μ and φ .

For v_{Ay} we have

$$v_{Ay} = v_y - \omega L \sin \varphi = v_0 - \frac{v_x}{\mu} - \frac{3v_x \sin \varphi}{\mu} (\sin \varphi - \mu \cos \varphi)$$

and also

$$N = \frac{m\dot{v}_x}{\mu}$$

from which

$$E_N = \int_0^{v_x} N v_{Ay} dt = \int_0^{v_x} \frac{m}{\mu^2} (\mu v_0 - v_x (1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))) dv_x$$

or

$$E_N = \frac{m}{\mu^2} \left(\mu v_0 v_x - \frac{v_x^2 (1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))}{2} \right)$$

Maximum value W be achieved for $v_{Ay} = 0$, thus be in this moment

$$v_x = \frac{\mu v_0}{1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi)}$$

from which

$$W = \frac{m v_0^2}{2(1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))}$$

Answer: Answer:

$$W = \frac{m v_0^2}{2(1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))}$$

C3

1.40 pts

Find v_{xto} after the hit. Express your answer in terms of v_0 , μ , φ and k .

At the end of the strike

$$E_N = E_{\text{def}} = W(1 - k) = \frac{m v_0^2 (1 - k)}{2(1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))}$$

from which

$$\frac{m}{\mu^2} \left(\mu v_0 v_x - \frac{v_x^2 (1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))}{2} \right) = \frac{m v_0^2 (1 - k)}{2(1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))}$$

or

$$v_x^2 - \frac{2\mu v_0 v_x}{1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi)} + \frac{\mu^2 v_0^2 (1 - k)}{(1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))^2} = 0$$

Solve square equation

$$v_x = \frac{\mu v_0 (1 \pm \sqrt{k})}{(1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))}$$

Necessarily choose larger from root. Thus

$$v_{xto} = \frac{\mu v_0 (1 + \sqrt{k})}{(1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))}$$

Answer: Answer:

$$v_{xto} = \frac{\mu v_0 (1 + \sqrt{k})}{(1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))}$$

Find the work done by the friction force A_{mp} during the impact. Express your answer in terms of m , v_0 , μ , φ and k .

For the work of the friction force we have

$$A_{\text{Tp}} = \int F_x v_{Ax} dt = \int_0^{v_x} m v_x \left(1 - \frac{3 \cos \varphi}{\mu} (\sin \varphi - \mu \cos \varphi) \right) dv_x$$

or

$$A_{\text{Tp}} = \frac{m v_x^2}{2} \left(1 - \frac{3 \cos \varphi}{\mu} (\sin \varphi - \mu \cos \varphi) \right)$$

Substitute find value v_x

$$A_{\text{Tp}} = \frac{\mu m v_0^2 (1 + \sqrt{k})^2 (\mu - 3 \cos \varphi (\sin \varphi - \mu \cos \varphi))}{2(1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))^2}$$

Answer: Answer:

$$A_{\text{Tp}} = \frac{\mu m v_0^2 (1 + \sqrt{k})^2 (\mu - 3 \cos \varphi (\sin \varphi - \mu \cos \varphi))}{2(1 + 3 \sin \varphi (\sin \varphi - \mu \cos \varphi))^2}$$